

Update on the Management of NSAID-Induced Lower Gastrointestinal Injury

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Why Lower GI Injury Matters in NSAID Users

Table 1 Capsule endoscopic prevalence of small-bowel mucosal breaks in subjects receiving NSAIDs or LDA

Author [Ref]	Year	N	Type of NSAIDs	Treatment period	Subject	Prevalence of mucosal breaks (%)
Graham et al. [9]	2005	21	Non-selective NSAIDs	> 3 months	Chronic user	62
Maiden et al. [16]	2005	40	Diclofenac (+ omeprazole)	2 weeks	Healthy volunteers	40
Goldstein et al. [17]	2005	111	Naproxen (+ omeprazole)	2 weeks	Healthy volunteers	55
		115	Celecoxib	2 weeks	Healthy volunteers	16
Goldstein et al. [18]	2007	112	Ibuprofen (+ omeprazole)	2 weeks	Healthy volunteers	26
		109	Celecoxib	2 weeks	Healthy volunteers	6
Maiden et al. [34]	2007	120	Non-selective NSAIDs	> 3 months	Chronic user	29
		40	Selective COX-2 inhibitors	> 3 months	Chronic user	22
Sugimori et al. [10]	2008	16	Non-selective NSAIDs	> 1 year	Chronic user	81
Hawkey et al. [30]	2008	45	Naproxen (+ omeprazole)	16 days	Healthy volunteers	78
		47	Lumiracoxib	16 days	Healthy volunteers	28
Fujimori et al. [23]	2009	15	Diclofenac (+ omeprazole)	2 weeks	Healthy volunteers	53
Maehata et al. [31]	2012	14	Celecoxib (+ omeprazole)	2 weeks	Healthy volunteers	43
		15	Meloxicam (+ omeprazole)	2 weeks	Healthy volunteers	27
Watanabe et al. [27]	2013	87	Non-selective NSAIDs	> 3 months	Chronic user	54
		21	Celecoxib	> 3 months	Chronic user	48
Watanabe et al. [37]	2008	11	Enteric-coated LDA (+ PPIs)	> 3 months	Chronic user with PUD	91
Sumecul et al. [38]	2009	20	Enteric-coated LDA (+ esomeprazole)	2 weeks	Healthy volunteers	20
Endo et al. [39]	2009	22	LDA	> 3 months	Chronic user with OGIB	46*
Endo et al. [40]	2009	10	Enteric-coated LDA	2 weeks	Healthy volunteers	30
Hara et al. [41]	2018	45	LDA	> 2 weeks	Chronic user	51

NSAID non-steroidal anti-inflammatory drug, LDA low-dose aspirin, COX cyclooxygenase, PPI proton pump inhibitor, PUD peptic ulcer disease, OGIB obscure gastrointestinal bleeding

*Prevalence of small-bowel ulcer

Lower GI events menyumbang 40% dari seluruh serious NSAID-related GI events

Studi capsule endoscopy klasik, cedera mukosa usus halus ditemukan pada 71% pengguna NSAID kronik versus 10% kontrol

NSAID-Induced Lower GI Injury

NSAID-induced lower GI injury → cedera pada small bowel dan kolon di distal duodenum, terutama enteropathy pada usus halus:

- Peningkatan permeabilitas
- Inflamasi mukosa
- Erosi
- Ulkus
- Perdarahan samar / nyata
- Protein loss
- Diaphragm-like stricture

Klinis bisa silent enteropathy (often missed) atau adanya penemuan endoskopi saja tapi bisa juga samapi dengan komplikasi:

- Anemia
- Hypoalbuminemia
- Perforasi
- Obstruksi
- Perdarahan

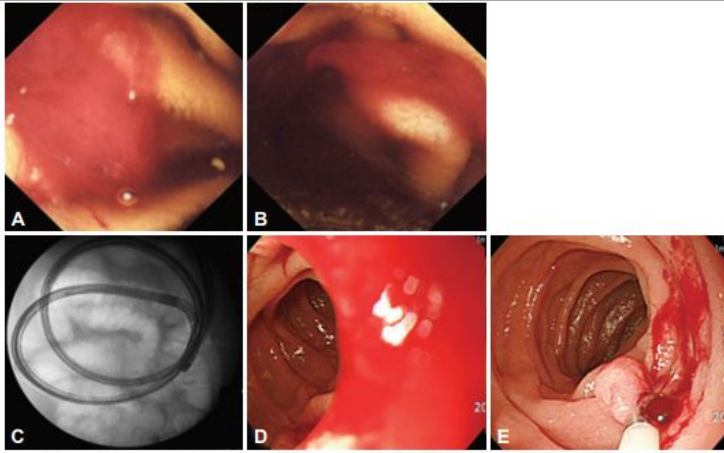


Fig. 2. A 40-year-old man with recurrent obscure gastrointestinal bleeding (Hb=5.0 mg/dL). He had intermittently taken non-steroidal anti-inflammatory drugs for intractable back pain. Capsule endoscopic finding (A, B) showed ongoing mid-jejunal bleeding without definite mucosal lesion and this bleeding was controlled by clipping through single balloon endoscopy (C, D, E).

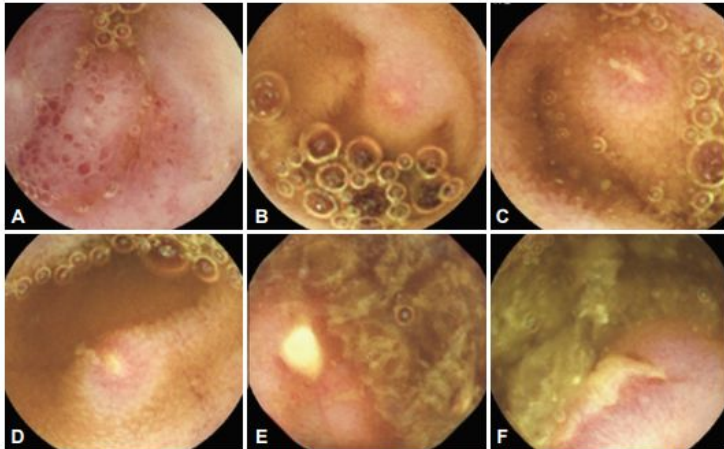
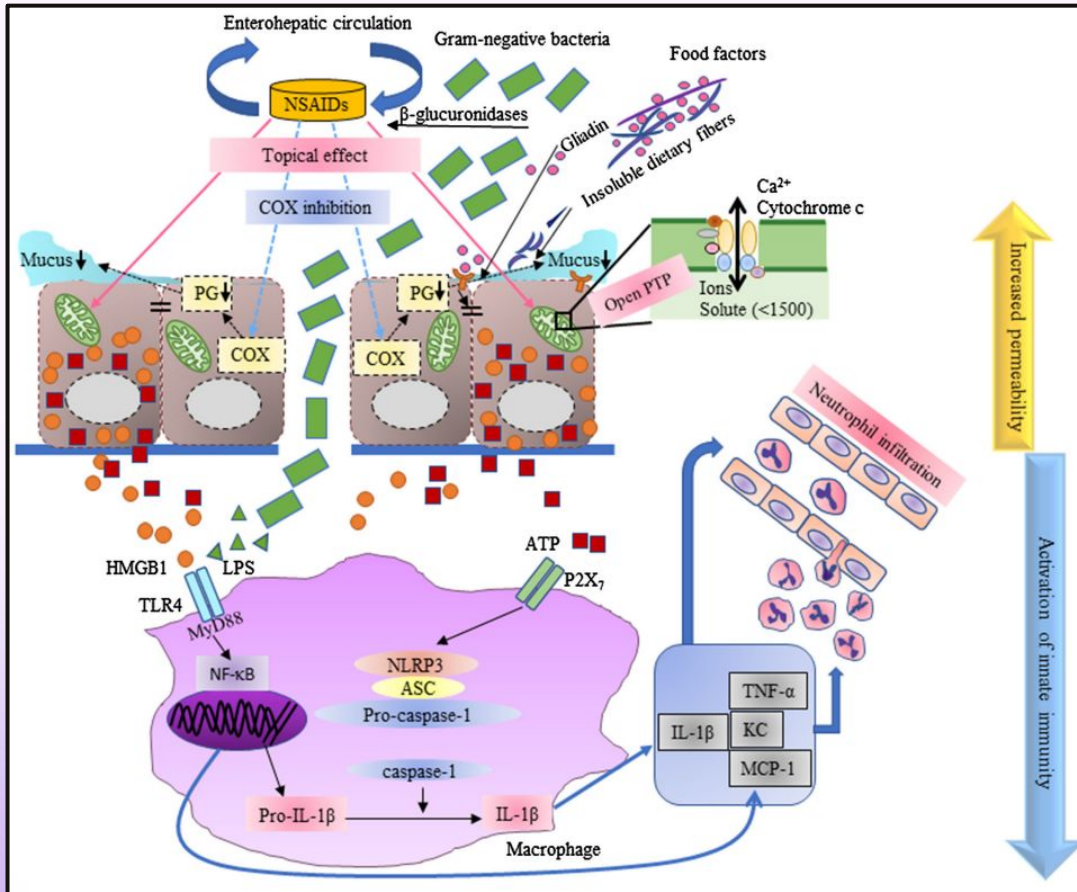


Fig. 3. Capsule endoscopic findings of non-steroidal anti-inflammatory drug enteropathy. There were denuded areas (A), erosions (B, C, D), and multiple variously-shaped well-demarcated ulcers (E, F).

Pathogenesis of Lower GI Injury



NSAID → epithelial injury + inhibisi COX → prostaglandin (PG) turun → lapisan mukus melemah + pertahanan mukosa menurun

Kerusakan enterosit/mitokondria membuka permeability transition pore (PTP) → pelepasan cytochrome c, gangguan ion (termasuk Ca²⁺) → permeabilitas mukosa meningkat → barrier usus menjadi bocor

Faktor luminal memperberat cedera:

- Enterohepatic circulation perpanjang paparan NSAID
- Bakteri Gram-negatif
- B-glucuronidase
- Faktor makanan (gliadin, insoluble dietary fibers)

Produk bakteri & sinyal bahaya masuk ke mukosa:

- LPS dan HMGB1 mengaktifkan jalur TLR4-MyD88-NF-κB
- ATP mengaktifkan P2X₇ dan inflammasom NLRP3/ASC/caspase-1

Terjadi respons inflamasi bawaan:

- Terbentuk IL-1β + TNF-α, KC, MCP-1 → memicu rekrutmen neutrofil → memperkuat inflamasi lokal

Risk Factors

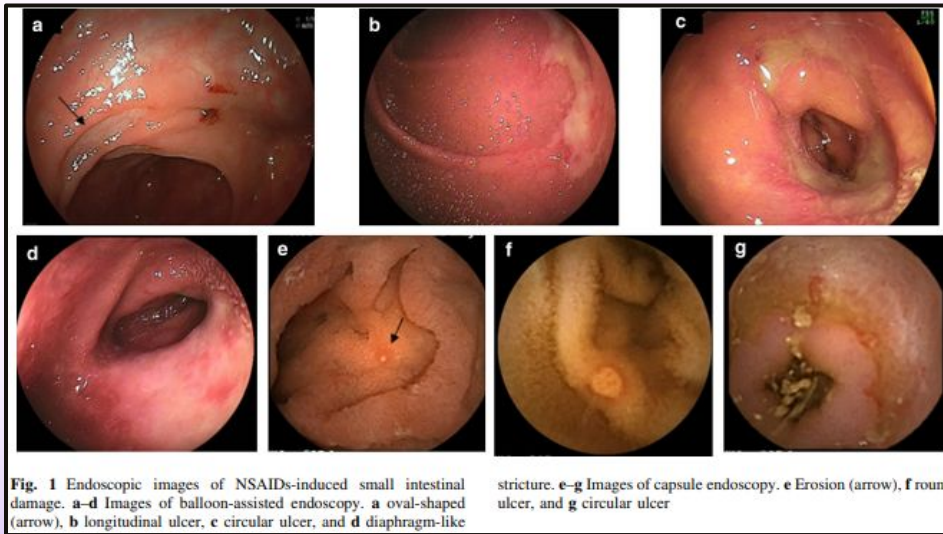
Table 1. Patients at increased risk for NSAID GI toxicity	
<i>High risk</i>	
1.	History of a previously complicated ulcer, especially recent
2.	Multiple (>2) risk factors
<i>Moderate risk (1–2 risk factors)</i>	
1.	Age >65 years
2.	High dose NSAID therapy
3.	A previous history of uncomplicated ulcer
4.	Concurrent use of aspirin (including low dose) corticosteroids or anticoagulants
<i>Low risk</i>	
1.	No risk factors
<i>H. pylori</i> is an independent and additive risk factor and needs to be addressed separately (see text and recommendations).	

Faktor risiko GI klasik:

- Usia lanjut
- Dosis tinggi
- Riwayat ulkus
- Penggunaan bersamaan aspirin
- Antikoagulam
- Kortikosteroid

Komorbid smoking, warfarin, dan faktor hostnya juga berhubungan dengan enteropathy

Clinical Presentation



- Occult bleeding / iron deficiency anemia tanpa sumber jelas pada pasien pengguna NSAID → curiga enteropathy
- Hypoalbuminemia / protein-losing enteropathy → hati ginjal tidak indikasikan penurunan albumin
- Keluhan samar: nyeri perut intermiten, kembung, diare, mual, penurunan toleransi makan, atau fatigue karena anemia
- Severe case:
 - Overt bleeding
 - Perforasi
 - Obstruksi
 - Diaphragm disease pada penggunaan NSAID jangka panjang

Diagnostic Approach

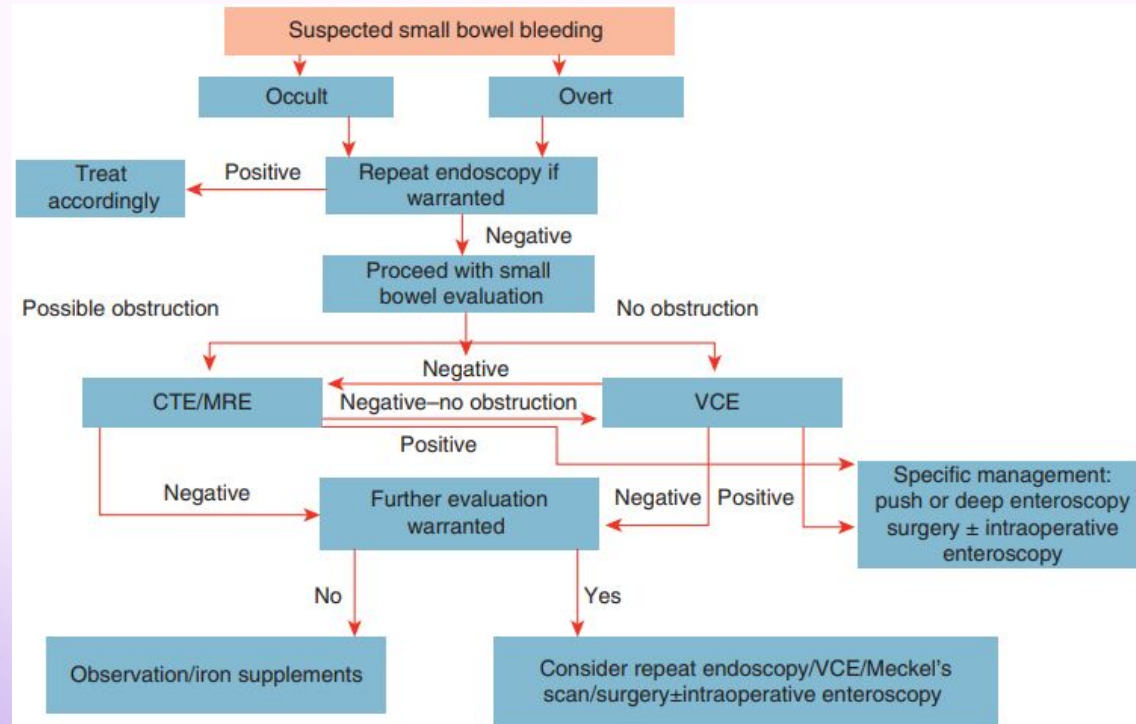


Figure 1. Algorithm for suspected small bowel bleeding. CTE, computed tomographic enterography; MRE, magnetic resonance enterography; VCE, video capsule endoscopy.

Konfirmasi indikasi NSAID dan menyingkirkan sumber perdarahan dari upper GI serta kolon

EGD/kolonoskopi tidak menemukan indikasi → video capsule endoscopy modalitas lini awal untuk evaluasi small bowel

CT/MR enterography dipertimbangkan → jika curiga stenosis, obstruksi, Crohn's disease, riwayat operasi, atau kapsul berisiko tertahan

Deep enteroscopy dilakukan bila perlu biopsi, hemostasis

Perdarahan aktif berat → CTA/angiografi

Diagnosis banding: Crohn's disease & penyebab ulkus/perdarahan usus halus laji

Management

Reassess need for NSAID → stop jika memungkinkan / ubahn dosis terendah, durasi tersingkat

COX-2 selective NSAID / NSAID + PPI/misoprostol → belum tentu selesaikan risiko lower GI

Table 2. Summary of recommendations for prevention of NSAID-related ulcer complications

	Gastrointestinal risk ^a		
	Low	Moderate	High
Low CV risk	NSAID alone (the least ulcerogenic NSAID at the lowest effective dose)	NSAID+PPI/misoprostol	Alternative therapy if possible or COX-2 inhibitor+PPI/misoprostol
High CV risk ^b (low-dose aspirin required)	Naproxen + PPI/misoprostol	Naproxen + PPI/misoprostol	Avoid NSAIDs or COX-2 inhibitors. Use alternative therapy
^a Gastrointestinal risk is stratified into low (no risk factors), moderate (presence of one or two risk factors), and high (multiple risk factors, or previous ulcer complications, or concomitant use of corticosteroids or anticoagulants). ^b High CV risk is arbitrarily defined as the requirement for low-dose aspirin for prevention of serious CV events. All patients with a history of ulcers who require NSAIDs should be tested for <i>H. pylori</i> , and if the infection is present, eradication therapy should be given.			

Rebamipide

Rebamipide adalah obat mucosal protective / cytoprotective agent turunan quinolinone

Rebamipide dipasarkan di Jepang pada 1990 untuk terapi ulkus lambung

1994, indikasinya diperluas ke lesi mukosa lambung terkait gastritis

Rebamipide diposisikan sebagai mucosal protective option dengan evidence yang menjanjikan

Mechanism of Action Rebamipide

Rebamipide meningkatkan prostaglandin/PGE2, merangsang sekresi mukus, dan mendukung epithelial restitution → pertahanan mukosa yang turun karena NSAID membaik

Rebamipide mengaktifkan jalur AMPK, ERK1/2, p38 MAPK, dan Akt/mTOR → survival sel, diferensiasi epitel, dan perbaikan mukosa usus lebih baik

Rebamipide merestorasi tight junction seperti ZO-1 dan occludin → intestinal permeability yang meningkat pada NSAID enteropathy dapat ditekan

Rebamipide menekan jalur TLR4/NF-κB, NF-κB, dan JNK, → mediator inflamasi TNF-α, IL-1β, IL-8 dan aktivasi neutrofil menurun.

Rebamipide bekerja sebagai barrier protector + anti-inflammatory + mucosal healing agent → lebih sesuai untuk small bowel injury dibanding yang cuma menekan asam lambung

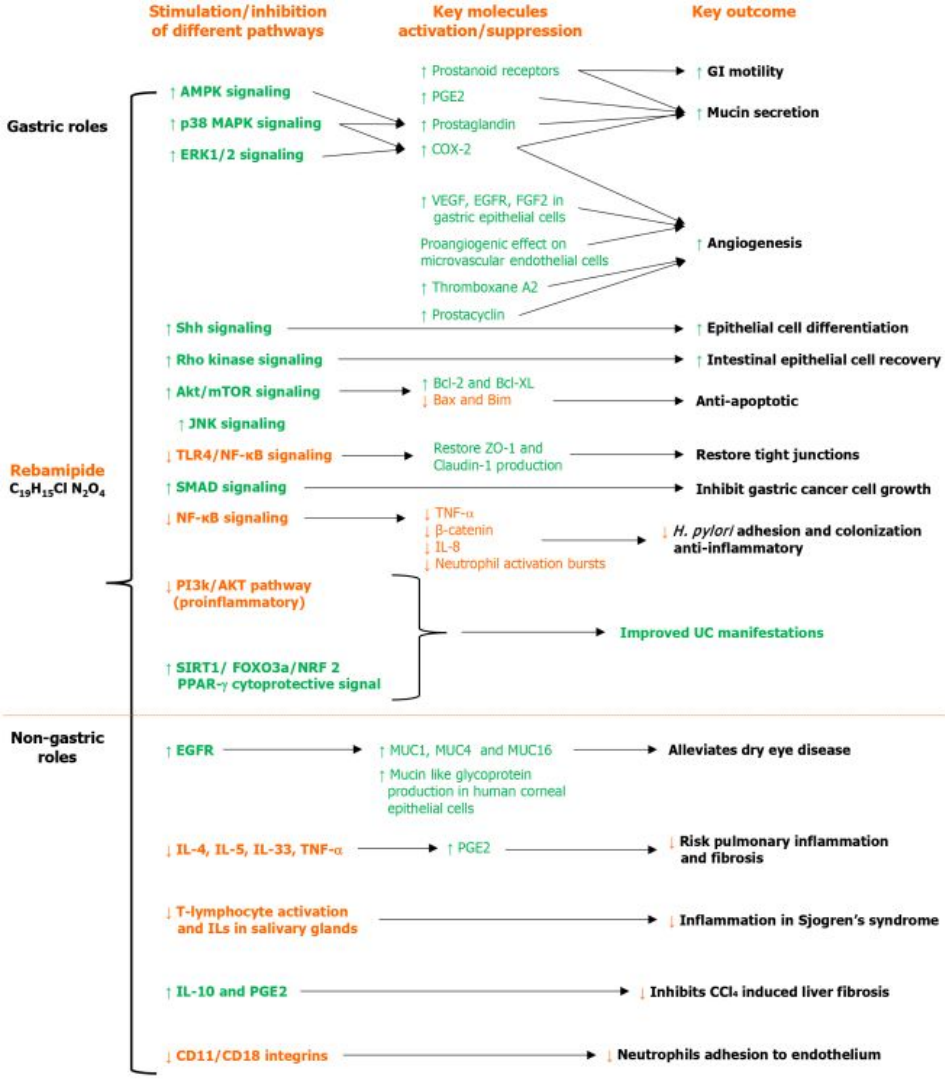


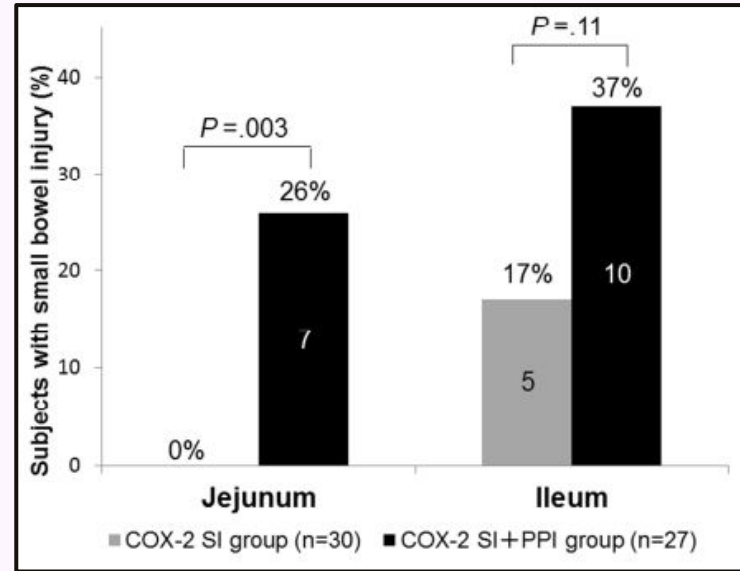
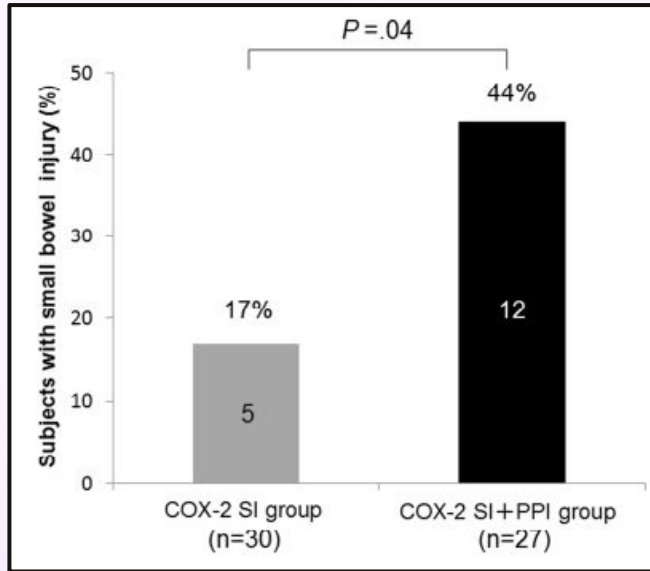
Figure 1 Molecular-level mechanisms of the action of rebamipide. AKT/mTOR: Protein kinase B/mammalian target of rapamycin; AMPK: Adenosine monophosphate-activated protein kinase; Bax: B-cell lymphoma 2 associated X; Bcl-2: B-cell lymphoma 2; Bcl-xL: B-cell lymphoma extra-large; Bim: B-cell lymphoma 2 interacting mediator of cell death; CCl₄: Carbon tetrachloride; CD: Cluster of differentiation; COX-2: Cyclooxygenase-2; EGFR: Epidermal growth factor receptor; ERK1/2: Extracellular signal-regulated protein kinase 1 and 2; FGF2: Fibroblast growth factor receptor-2; GI: Gastrointestinal; *H. Pylori*: *Helicobacter pylori*; IL: interleukin; JNK: Jun N-terminal kinase; MUC: Mucin; NF-κB: Nuclear factor kappa B; PGE2: Prostaglandin E2; PI3K/AKT: Phosphoinositide 3-kinase/protein kinase B; PPAR-γ: Proliferator-activated receptor gamma; P38MAPK: P38 mitogen-activated protein kinase; Shh: Sonic hedgehog; SIRT1/FoxO3a/NRF2: Silent information regulator-1/forkhead box O3a/nuclear factor erythroid 2-related factor 2; SMAD: Suppressor of mothers against decapentaplegic; TLR4/NF-κB: Toll-like receptor 4/nuclear factor kappa B; TNF-α: Tumour necrosis factor alpha; UC: Ulcerative colitis; VEGF: Vascular endothelial growth factor; ZO-1: Zonula-occludens.

PPIs Are Not a Complete Lower GI Solution

PPI efektif untuk upper GI protection, tetapi tidak melindungi usus halus distal dari NSAID injury

Co-therapy NSAID + PPI terbukti mengurangi gastroduodenal injury, namun enteropathy tetap dapat terjadi

Kejadian small-bowel injury dilaporkan lebih tinggi pada celecoxib + rabeprazole dibanding celecoxib + placebo



Konsep whole-GI mucosal protection relevan pasien yang tetap harus melanjutkan NSAID

Rebamipide pada NSAID-Induced Small Bowel Damage

Metode: systematic review dan meta-analysis

Populasi: 15 RCT dengan total 965 pengguna NSAID + cedera mukosa GI, termasuk kelompok small bowel injury/lower GI

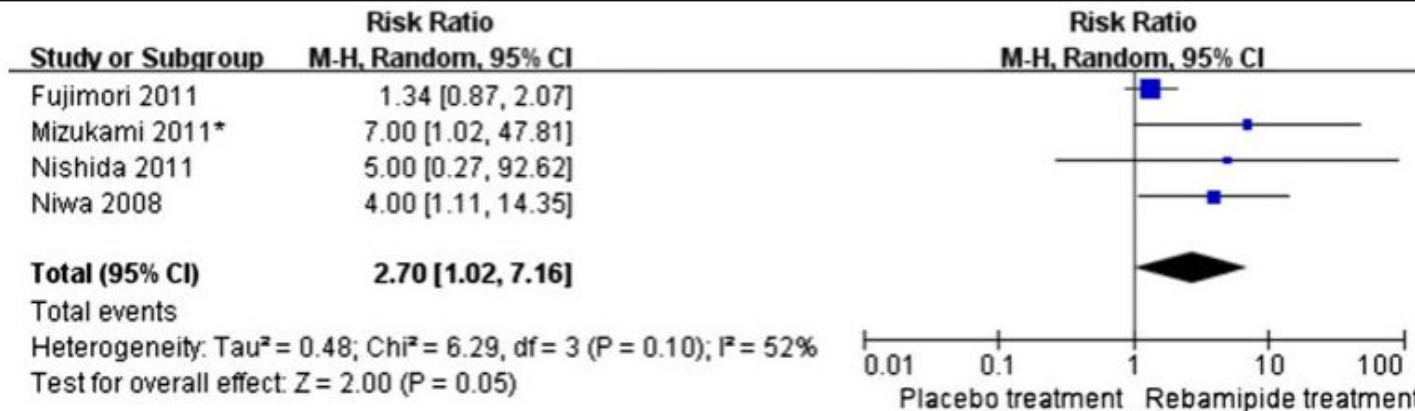
Intervensi: rebamipide vs placebo / terapi komparator, 100 mg 3x sehari

Hasil:

- Small bowel damage: rebamipide menunjukkan perbaikan signifikan vs placebo dengan RR 2,70 (95% CI 1,02–7,16; $p=0,045$)
- Tidak ada serious adverse event yang dilaporkan

Kesimpulan: Rebamipide mendukung proteksi NSAID-induced gastroenteropathy terutama pada small bowel damage

Fig. 5 Rebamipide potentially protects against NSAID-induced small-bowel damage. *Data evaluating short-term treatment effect (1-week) are included



Rebamipide vs Lansoprazole pada Long-Term NSAID Enteropathy

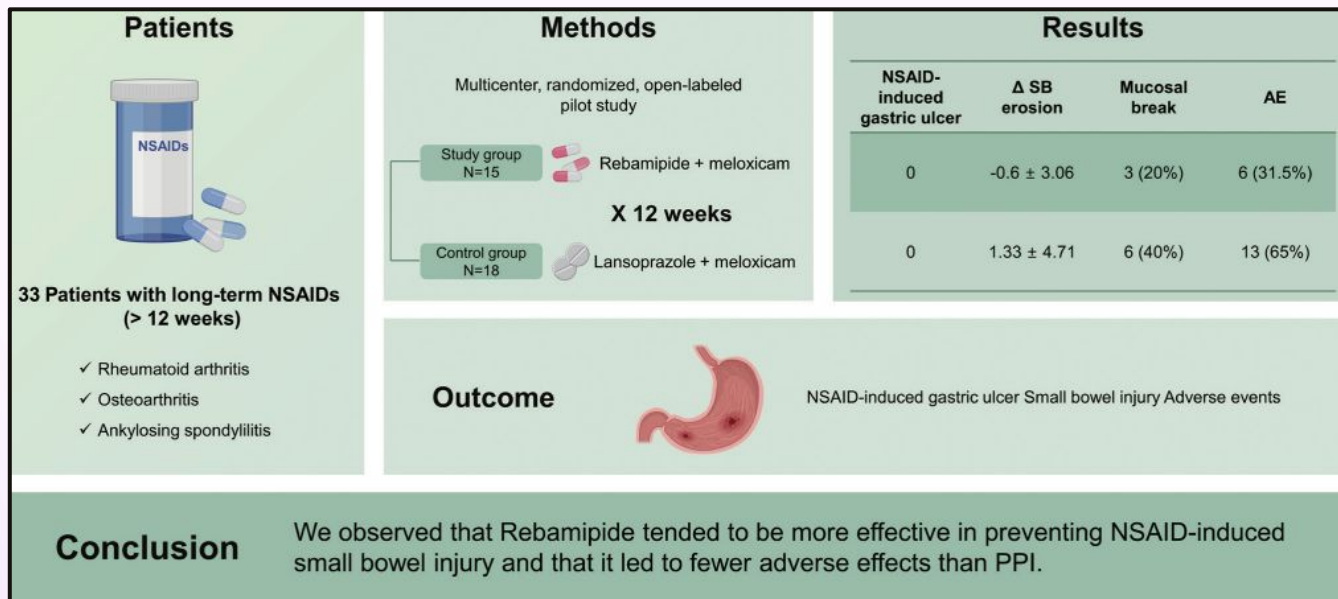
Metode: multicenter randomized open-label pilot study

Populasi: 33 pasien dewasa rerate usia 55 tahun dengan rheumatoid arthritis, osteoarthritis, ankylosing spondylitis, dan penyakit muskuloskeletal lain yang memerlukan NSAID kontinu

Intervensi: Rebamipide 100 mg tiga kali sehari vs lansoprazole 15 mg sekali sehari, diberikan selama 12 minggu bersama NSAID

- Small bowel mucosal breaks ditemukan di 20% kelompok rebamipide vs 40% kontrol ($p=0,427$)
- Erosions/ulcers cenderung lebih baik pada rebamipide
- Efek samping lebih rendah rebamipide (31,58% vs 65%; $p=0,036$)

Kesimpulan: Rebamipide mendukung proteksi small bowel pada penggunaan NSAID long term



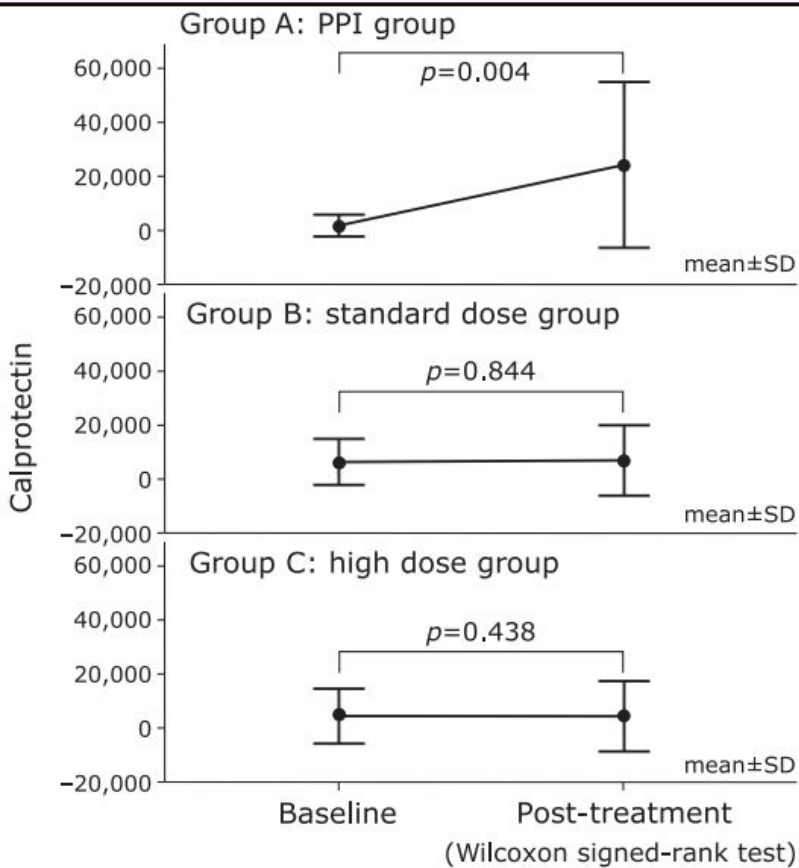


Fig. 5. Fecal calprotectin levels before and after treatment. A significant increase in the fecal calprotectin levels was observed between baseline and post-treatment in Group A ($p = 0.004$). Conversely, there were no significant differences in the calprotectin levels between the baseline and post-treatment in Group B ($p = 0.844$) or Group C ($p = 0.438$).

Rebamipide pada Low-Dose Aspirin-Induced Small Intestinal Injury

Metode: prospective randomized dose-finding trial

Populasi: 45 volunteer dewasa sehat, usia 20–65 tahun

Intervensi: seluruh mendapat enteric-coated aspirin 100 mg/hari selama 2 minggu

+ omeprazole 10 mg/hari vs rebamipide 300 mg/hari vs rebamipide 900 mg/hari

Hasil:

- Fecal Calprotectin meningkat signifikan pada kelompok omeprazole ($p=0,004$)
- Fecal Calprotectin tidak meningkat pada kelompok rebamipide 300 mg ($p=0,844$) dan 900 mg ($p=0,438$)
- EGD, capsule endoscopy, dan fecal occult blood tidak berbeda signifikan antar kelompok

Kesimpulan: Rebamipide menekan inflammation-related lower GI injury signal pada aspirin enteropathy

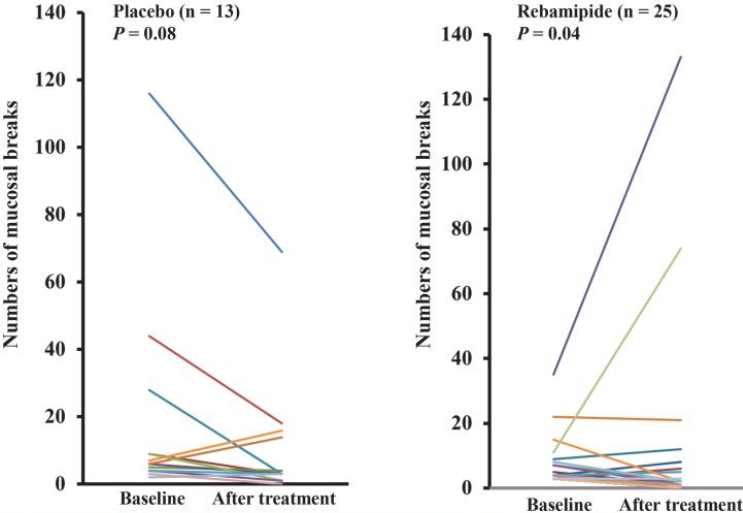


Fig 3. Changes in the number of small intestinal mucosal breaks from baseline to posttreatment. Results were analyzed by the Wilcoxon signed-rank test.

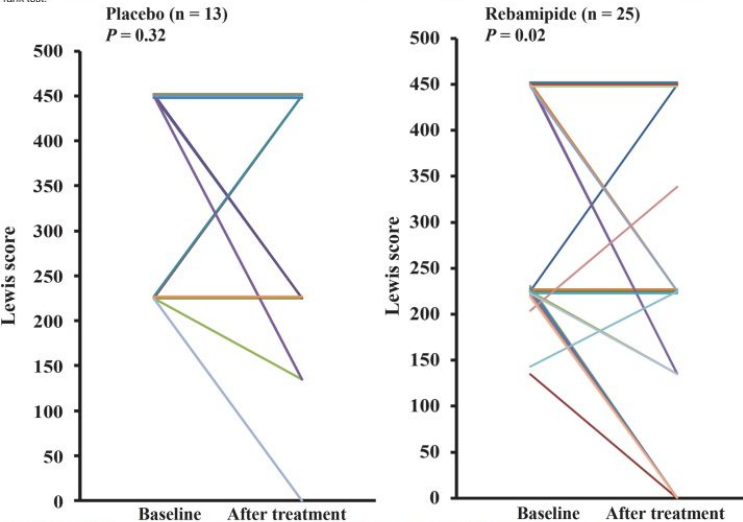


Fig 4. Changes in the severity of small intestinal mucosal breaks from baseline to posttreatment. Results were analyzed by McNemar test.

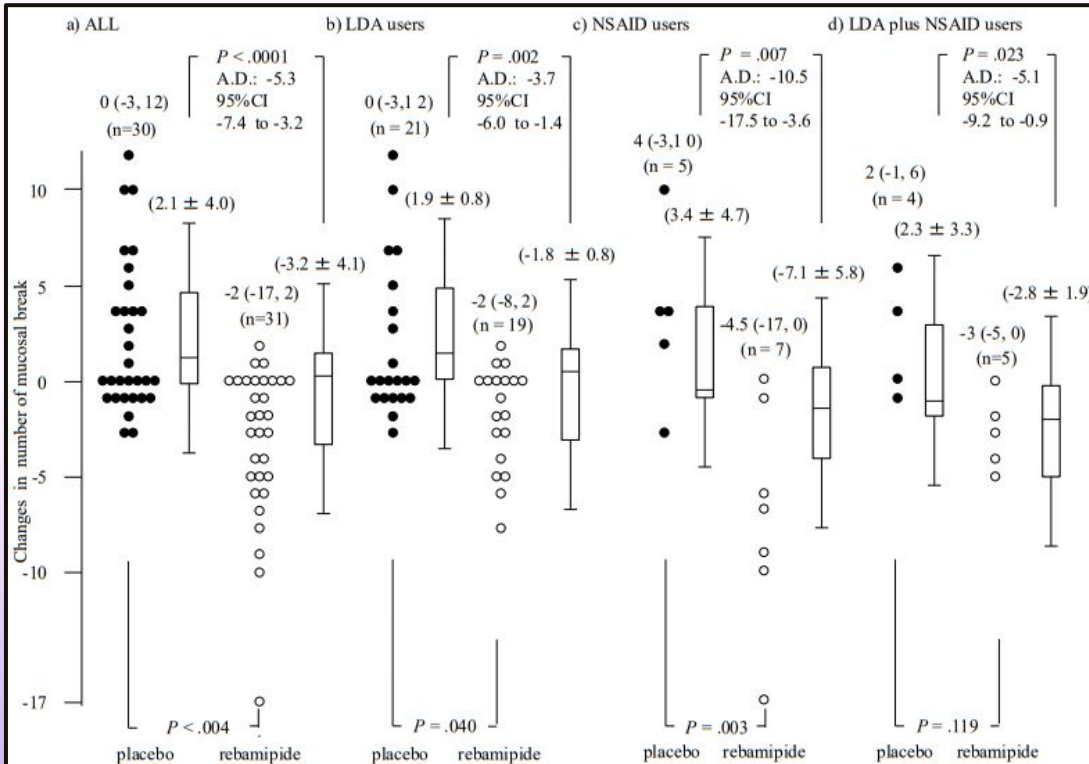
High-Dose Rebamipide pada Established Aspirin-Induced Enteropathy

Metode: multicenter, randomized, double-blind, placebo-controlled trial
 Populasi: 38 dewasa ≥ 20 tahun dengan enteric-coated aspirin 100 mg/hari > 3 bulan + > 3 mucosal breaks pada capsule endoscopy
 Intervensi: Reebamipide 300 mg tiga kali sehari (900 mg/hari) vs placebo selama 8 minggu
 Hasil:

- Jumlah small bowel mucosal breaks menurun bermakna pada kelompok rebamipide ($p=0,046$)
- Lewis score membaik signifikan ($p=0,02$)
- Complete healing terjadi pada 32,0% vs 7,7%

Kesimpulan: Rebamipide memperbaiki lesi small bowel secara signifikan pada pasien dengan aspirin-induced enteropathy

Rebamipide untuk Healing Chronic LDA/NSAID Small Bowel Injury



Metode: randomized, double-blinded, placebo-controlled, multicenter trial
 Populasi: 61 pengguna kronik low-dose aspirin dan/atau NSAID >3 bulan dengan small bowel injury pada capsule endoscopy
 Intervensi: Rebamipide 100 mg tiga kali sehari selama 4 minggu vs placebo
 Hasil:

- Jumlah mucosal breaks lebih baik pada rebamipide P<0,004; erosi turun secara signifikan(P<0,0001)
- Ulkus turun secara signifikan (P=0,024)
- Complete remission pada 37,5% vs 5,3% (P=0,018)
- Total protein membaik secara signifikan (P=0,0005)

Kesimpulan: Rebamipide memiliki healing effect secara signifikan pada small bowel lesions pengguna LDA/NSAID

Rebamipide Mencegah Ileal Mucosal Breaks akibat Aspirin

Table 1 No. of healthy subjects presenting small bowel injuries following a 4 wk regimen of low dose aspirin in the placebo group and evaluation of preventive efficacy ($n = 11$)

	Jejunum			Ileum		
	Pre-test	1 wk	4 wk	Pre-test	1 wk	4 wk
Placebo group						
Multiple erosion (n)	0	1	1	0	6 ^b	4 ^a
Ulcer (n)	0	0	0	0	0	3
Mucosal break (n)	0	1	1	0	6 ^b	7 ^b
Rebamipide group						
Mucosal break (n)	0	0	1	0	1 ^c	2 ^c

Multiple erosion (n): No. of subjects with multiple erosions in jejunum or ileum; Ulcer (n): No. of subjects with ulcer in jejunum or ileum; Mucosal break (n): No. of subjects with multiple erosion and/or ulcer in jejunum or ileum. Statistical analysis was performed by Mann Whitney's U test, compared to before administration *vs* 1 wk and 4 wk. ^a $P < 0.05$; ^b $P < 0.01$; ^c $P < 0.05$ compared with placebo and rebamipide group.

Metode: single-center, randomized, double-blind, cross-over, placebo-controlled study

Populasi: 11 pria dewasa sehat, usia 20–65 tahun

Intervensi: Rebamipide 300 mg 3x sehari / placebo, + aspirin 100 mg 1x sehari & omeprazole 20 mg 1x sehari, 4 minggu, lalu cross-over setelah washout

Hasil:

- Rebamipide secara signifikan mencegah ileal mucosal breaks dibanding placebo di minggu 1 ($p=0,0173$) dan minggu 4 ($p=0,0266$)
- Periode placebo, lesi ileum meningkat setelah pemberian aspirin

Kesimpulan: Rebamipide mencegah small bowel injury terutama di ileum pada aspirin-induced enteropathy

Table 1 The evaluation of area under the curve and time intensity curve by ultrasonography before and after taking low-dose aspirin ($n = 10$)

	Placebo			Rebamipide		
	A.D.	95% CI	P-value	A.D.	95% CI	P-value
AUC	-1102.5	-1980.3 to -224.7	0.0194	-152.7	-1604.2-641.6	0.8172
TIC	-148.0	-269.4 to -26.2	0.0225	28.3	-269.0-325.6	0.8343

A.D.: Absolute differences; AUC: Area under the curve, TIC: Time intensity curve.

Table 2 The evaluation of the preventive effect of rebamipide on low-dose aspirin-induced worsened small bowel injury compared with placebo (mean $\pm$ SD)

	Placebo	Rebamipide
Erosion	0.5 $\pm$ 2.7	0.0 $\pm$ 0.0
Petechiae	10.1 $\pm$ 53.6	-8.6 $\pm$ 15.4
Erythema	-1.0 $\pm$ 2.3	-0.6 $\pm$ 1.0

Table 3 Number of subjects with small intestinal mucosal breaks

	Placebo	Rebamipide
Mucosal breaks	2	0

No ulcer, bleeding or perforation was observed. There were 2 cases of mucosal break in the placebo group but no cases in the rebamipide group (Table 3). Two cases of erosions were localized in the ileum region (Figure 3).

Safety assessment

No adverse events were observed throughout the study period.

DISCUSSION

In the present study, treatment with low-dose ASA result-

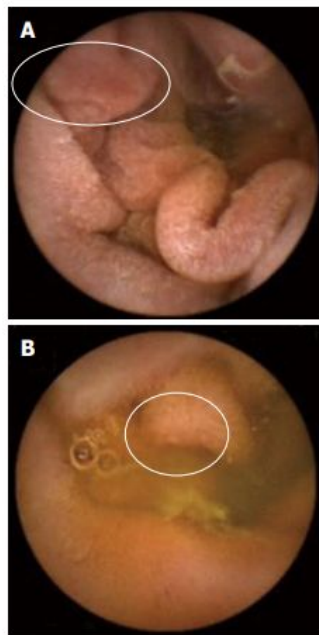


Figure 3 Image of capsule endoscopy. Low-dose aspirin induced mucosal breaks such as denuded areas and erosions were detected in the placebo group. A: Denuded area; B: Erosion.

Rebamipide Menjaga Small Bowel Blood Flow pada Aspirin Injury

Metode: randomized, double-blind, cross-over, placebo-controlled trial

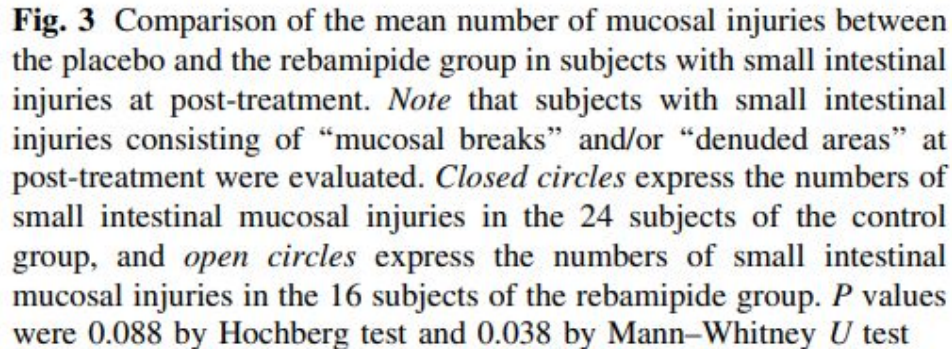
Populasi: 10 volunteer dewasa sehat, usia 20–50 tahun

Intervensi: low-dose aspirin dengan placebo vs low-dose aspirin dengan rebamipide selama 14 hari

Hasil:

- Penurunan AUC blood flow signifikan ($P=0,0194$) dan TIC peak ($P=0,0225$) pada plasebo
- Penurunan tidak signifikan pada rebamipide
- Capsule endoscopy menunjukkan mucosal breaks cuma pada placebo (2 vs 0 kasus)

Kesimpulan: Rebamipide mempertahankan perfusi mukosa small bowel + menurunkan kemunculan mucosal breaks



Metode: double-blind randomized controlled trial
Populasi: 72 pria dewasa sehat, 34 rebamipide – 38 kontrol/placebo pada cedera usus halus akibat diklofenak
Intervensi: Rebamipide 300 mg/hari vs placebo + diclofenac 75 mg/hari selama 2 minggu
Hasil:

- Jumlah rata-rata lesi mukosa per orang menurun dari $25,1 \pm 89,3$ menjadi $8,9 \pm 9,4$ pada pemberian rebamipdie $p=0,038$

Kesimpulan: Rebamipide mengurangi intensitas small bowel injury akibat diklofenak

Take Home Messages

NSAID-induced lower GI injury sering silent → curiga pada pasien dengan occult bleeding, anemia defisiensi besi, hypoalbuminemia

Proteksi upper GI $\neq$ proteksi lower GI: PPI saja tidak melindungi small bowel distal dan enteropathy tetap bisa terjadi

Kalau NSAID tetap harus dilanjutkan, butuh pendekatan whole-GI mucosal protection → bukan hanya acid suppression.

Rebamipide relevan karena bekerja sebagai mucosal protector: meningkatkan prostaglandin/mukus, memperbaiki barrier usus, dan menekan inflamasi mukosa

Jurnal jurnal terbaru menunjukkan rebamipide mendukung proteksi dan healing small bowel injury pada NSAID/LDA-induced enteropathy

Thank You